b.Develop an Android application demonstrating the use of Programming Resources (colors.xml, themes.xml, strings.xml, dimens.xml, drawable resources) and apply them in UI components.

Programming Resources

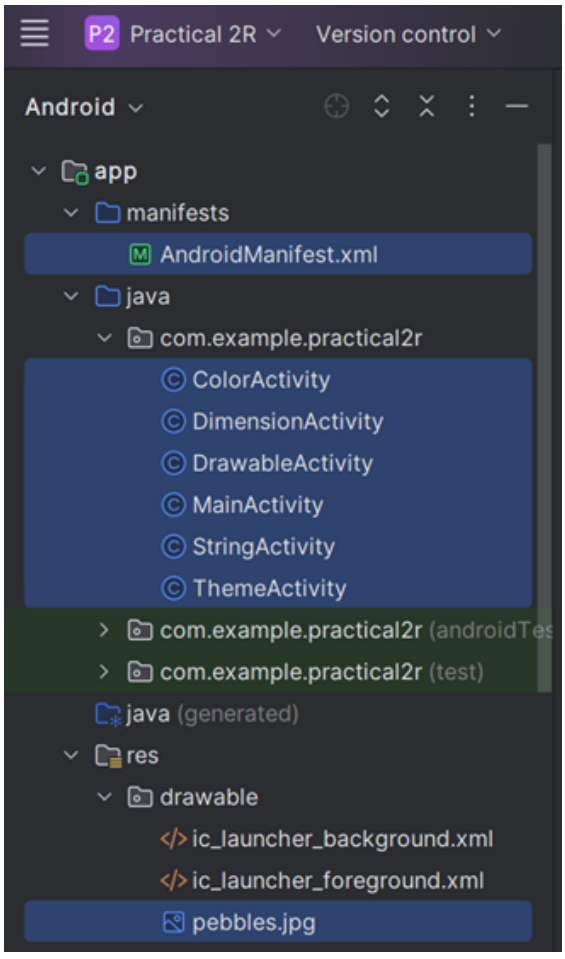
Android Resources: (Color, Theme, String, Drawable, Dimension, Image)

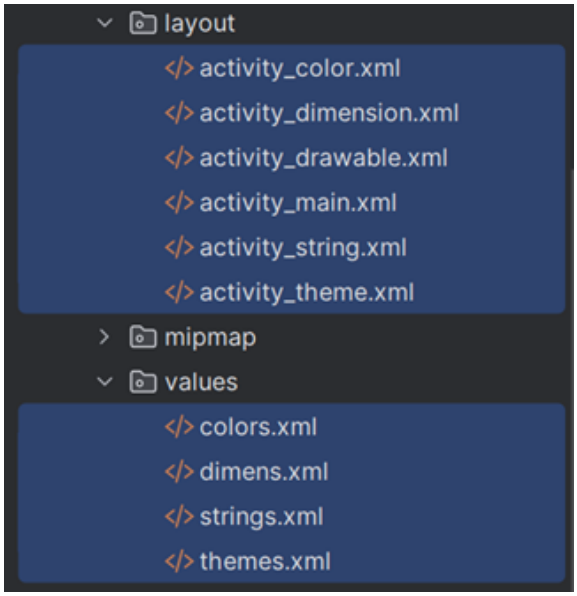
Project Name: Practical 2R

Template: Empty Views Activity

Language: Java

Project Architecture: 1 AndroidManifest.xml file, 6 Java Files,1 image in drawable image file 6 layout .xml files and 4 values .xml files: Total 18 files.





Filename: AndroidManifest.xml

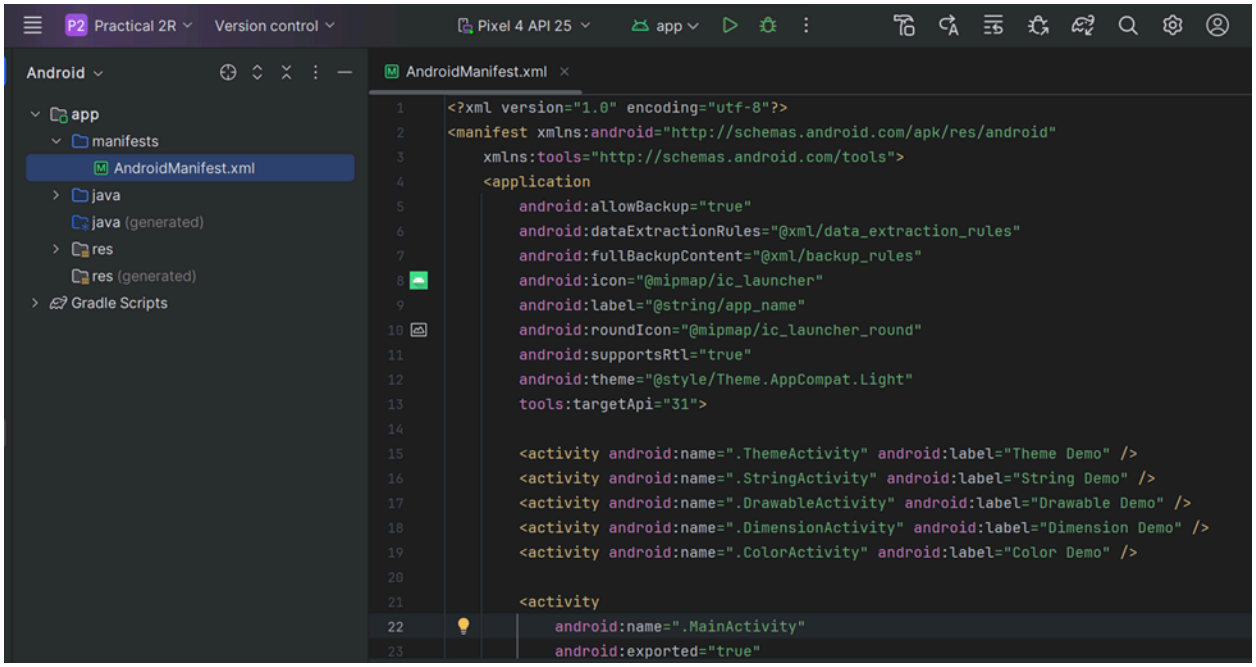
```
<?xml version="1.0" encoding="utf-8"?>
 <manifest xmlns:android="http://schemas.android.com/apk/res/android"
      xmlns:tools="http://schemas.android.com/tools">
      <application
      android:allowBackup="true"
      android:dataExtractionRules="@xml/data_extraction_rules"
      android:fullBackupContent="@xml/backup_rules"
      android:icon="@mipmap/ic_launcher"
      android:label="@string/app_name"
      android:roundIcon="@mipmap/ic_launcher_round"
      android:supportsRtl="true"
      android:theme="@style/Theme.AppCompat.Light"
      tools:targetApi="31">

      <activity android:name=".ThemeActivity" android:label="Theme Demo" />
      <activity android:name=".StringActivity" android:label="String Demo" />
      <activity android:name=".DrawableActivity" android:label="Drawable Demo"
/>
      <activity android:name=".DimensionActivity" android:label="Dimension
Demo" />
      <activity android:name=".ColorActivity" android:label="Color Demo" />

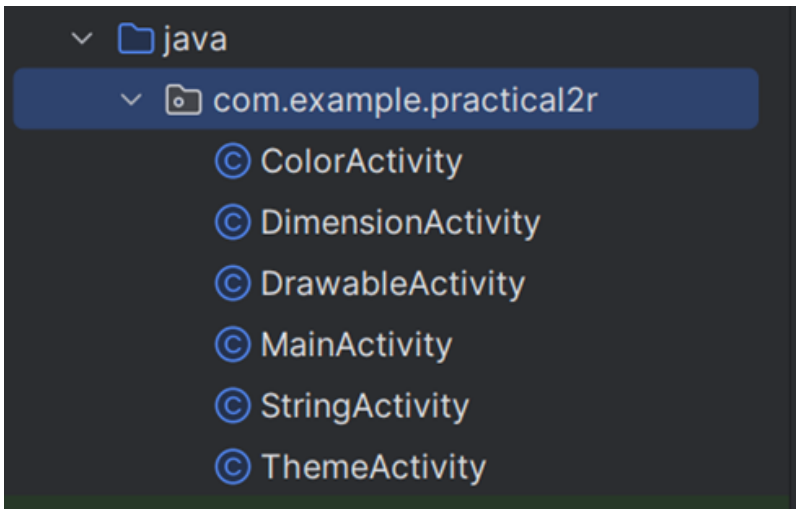
      <activity
            android:name=".MainActivity"
            android:exported="true"
            android:label="@string/app_name"
            android:theme="@style/Theme.AppCompat.Light">
            <intent-filter>
                  <action android:name="android.intent.action.MAIN" />
```

```
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        </application>
</manifest>
```

Path: app>manifests>AndroidManifest.xml



Total 6 .java files path



1.Filename: ColorActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class ColorActivity extends AppCompatActivity {
     @Override
     protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_color);
     }
}
```

2.Filename: DimensionActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class DimensionActivity extends AppCompatActivity {
     @Override
     protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_dimension);
     }
}
```

3.Filename: DrawableActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class DrawableActivity extends AppCompatActivity {
     @Override
     protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_drawable);
     }
}
```

4.Filename: MainActivity.java

```
package com.example.practical2r;

 import android.content.Intent;
 import android.os.Bundle;
 import android.view.View;
 import androidx.appcompat.app.AppCompatActivity;

 public class MainActivity extends AppCompatActivity {
      @Override
      protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      }
      public void openColorActivity(View view) {
      startActivity(new Intent(this, ColorActivity.class));
      }
      public void openThemeActivity(View view) {
      startActivity(new Intent(this, ThemeActivity.class));
      }
      public void openStringActivity(View view) {
      startActivity(new Intent(this, StringActivity.class));
      }
      public void openDrawableActivity(View view) {
      startActivity(new Intent(this, DrawableActivity.class));
      }
      public void openDimensionActivity(View view) {
      startActivity(new Intent(this, DimensionActivity.class));
      }
 }
```

5.Filename: StringActivity.java

```
package com.example.practical2r;

 import android.os.Bundle;
 import androidx.appcompat.app.AppCompatActivity;

 public class StringActivity extends AppCompatActivity {
      @Override
      protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_string);
      }
 }
```

6.Filename: ThemeActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class ThemeActivity extends AppCompatActivity {
     @Override
     protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_theme);
     }
}
```

Image File Path

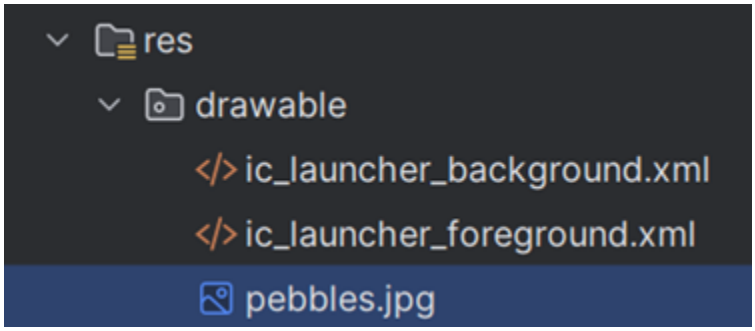
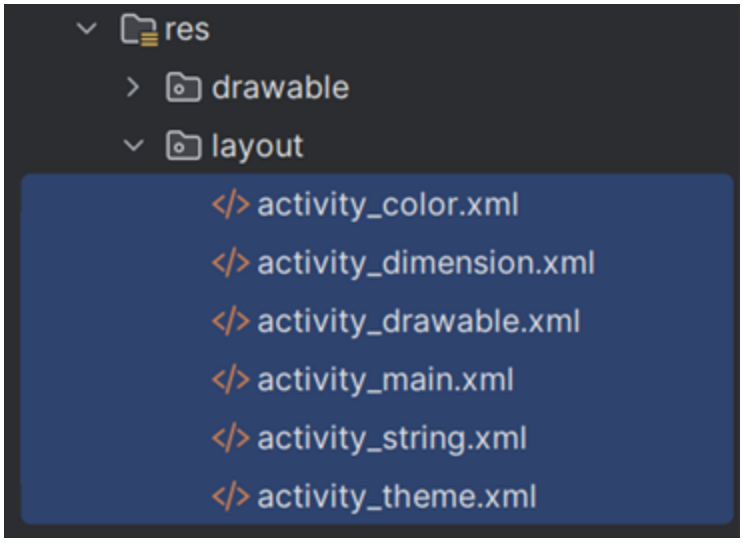


Image Detail: pebbles.jpg

(Copy your image file from your local folder and only copy and paste into drawable folder. Don't Drag and Drop). Image size must within KB size



Total 6 Layout .xml files path



1.Filename: activity_color.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:background="@color/colorRed">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Color Activity"
      android:textSize="24sp"
          android:layout_centerInParent="true"
      android:textColor="@color/white"/>
 </RelativeLayout>
```

2.Filename: activity_dimension.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:orientation="vertical"
      android:padding="16dp"
      android:gravity="center">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Dimension Activity"
      android:textSize="24sp"
```

```
      android:textColor="@android:color/black"/>
      <!-- Button demonstrating dimension usage -->
      <Button
         android:id="@+id/dimensionButton"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Click Me!"
      android:textSize="@dimen/button_text_size"
      android:padding="@dimen/button_padding"
      android:layout_marginTop="@dimen/button_margin_top"
      android:background="@color/btn_color"/>
 </LinearLayout>
```

3.Filename: activity_drawable.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:background="@drawable/pebbles">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Drawable Activity"
      android:textSize="24sp"
      android:layout_centerInParent="true"
      android:textColor="@android:color/white"/>
 </RelativeLayout>
```

4.Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <RelativeLayout
      xmlns:android="http://schemas.android.com/apk/res/android"
      xmlns:tools="http://schemas.android.com/tools"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      tools:context=".MainActivity"
      android:background="@drawable/pebbles">
      <Button
      android:id="@+id/btnColor"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Color Demo"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="50dp"
      android:onClick="openColorActivity"/>
      <Button
      android:id="@+id/btnTheme"
```

```
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Theme Demo"
      android:layout_below="@id/btnColor"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="20dp"
      android:onClick="openThemeActivity"/>
      <Button
      android:id="@+id/btnString"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="String Demo"
      android:layout_below="@id/btnTheme"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="20dp"
         android:onClick="openStringActivity"/>

      <Button
      android:id="@+id/btnDrawable"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Drawable Demo"
      android:layout_below="@id/btnString"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="20dp"
      android:onClick="openDrawableActivity"/>
      <Button
      android:id="@+id/btnDimension"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Dimension Demo"
      android:layout_below="@id/btnDrawable"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="20dp"
      android:onClick="openDimensionActivity"/>
</RelativeLayout>
```

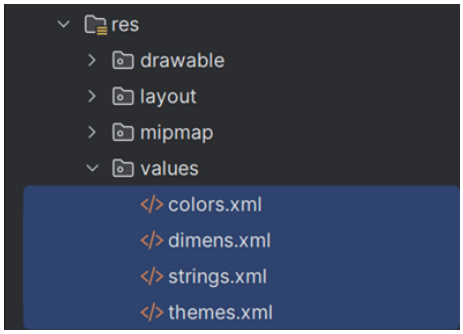
5.Filename: activity_string.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:orientation="vertical"
      android:gravity="center"
      android:background="@color/colorPrimaryDark">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="@string/Welcome_Message"
      android:textSize="24sp"
      android:textColor="@android:color/white"/>
 </LinearLayout>
```

6.Filename: activity_theme.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:background="@android:color/holo_blue_bright">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="You got this sky blue theme"
      android:textSize="24sp"
      android:layout_centerInParent="true"
      android:textColor="@android:color/black"/>
 </RelativeLayout>
```

Total 4 values .xml files



1.Filename: colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <color name="purple_200">#FFBB86FC</color>
      <color name="purple_500">#FF6200EE</color>
      <color name="purple_700">#FF3700B3</color>
      <color name="teal_200">#9ADA03</color>
      <color name="teal_700">#FF018786</color>
      <color name="black">#FF000000</color>
      <color name="white">#FFFFFFFF</color>
      <color name="btn_color">#FFC49662</color>
      <color name="colorRed">#E91E63</color>
      <color name="colorBlue">#0000FF</color>
      <color name="colorPrimary">#6200EE</color>
      <color name="colorPrimaryDark">#3700B3</color>
      <color name="colorAccent">#03DAC5</color>
 </resources>
```

2.Filename: dimens.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <dimen name="button_margin_top">24dp</dimen>
      <dimen name="button_padding">16dp</dimen>
      <dimen name="button_text_size">18sp</dimen>
 </resources>
```

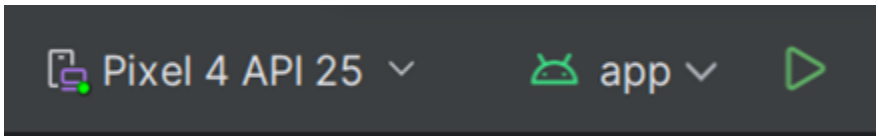
3.Filename: strings.xml

```
<resources>
      <string name="app_name">Practical 2R</string>
      <string name="button_name">Click on me</string>
      <string name="ParaHead">Welcome to Practical 2R</string>
      <string name="Description">This is a string resource demo.</string>
      <string name="Welcome_Message">Welcome to Android Studio. This is your
Practical 2 Resources.</string>
 </resources>
```

4.Filename: themes.xml

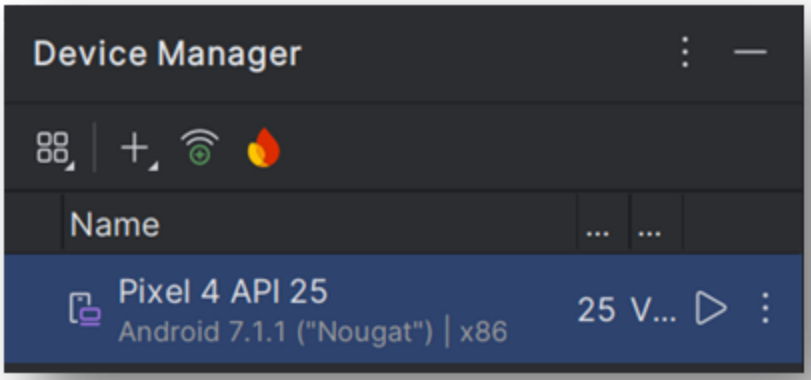
```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <style name="AppTheme" parent="Theme.AppCompat.Light.DarkActionBar">
        <item name="colorPrimary">@color/colorPrimary</item>
      <item name="colorPrimaryDark">@color/colorPrimaryDark</item>
      <item name="colorAccent">@color/colorAccent</item>
      </style>
      <style name="BlueTheme" parent="Theme.AppCompat.Light.DarkActionBar">
      <item name="colorPrimary">@color/colorPrimary</item>
      <item name="colorPrimaryDark">@color/colorPrimaryDark</item>
      <item name="colorAccent">@color/colorAccent</item>
      <item name="android:background">#00F0EF</item>
      <item name="android:textColor">#000000</item>
      </style>
 </resources>
```

Above the Android Studio near Main Menu Click the Play button to Run the Program



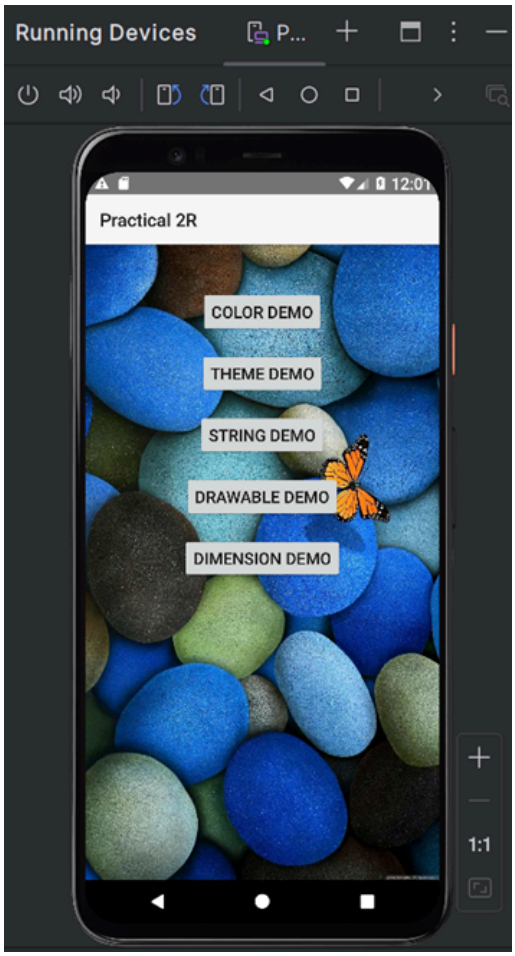
Recommended compulsory AVD Emulator: Pixel 4 API 25 with version 7.1.1 Nougat

Play Emulator

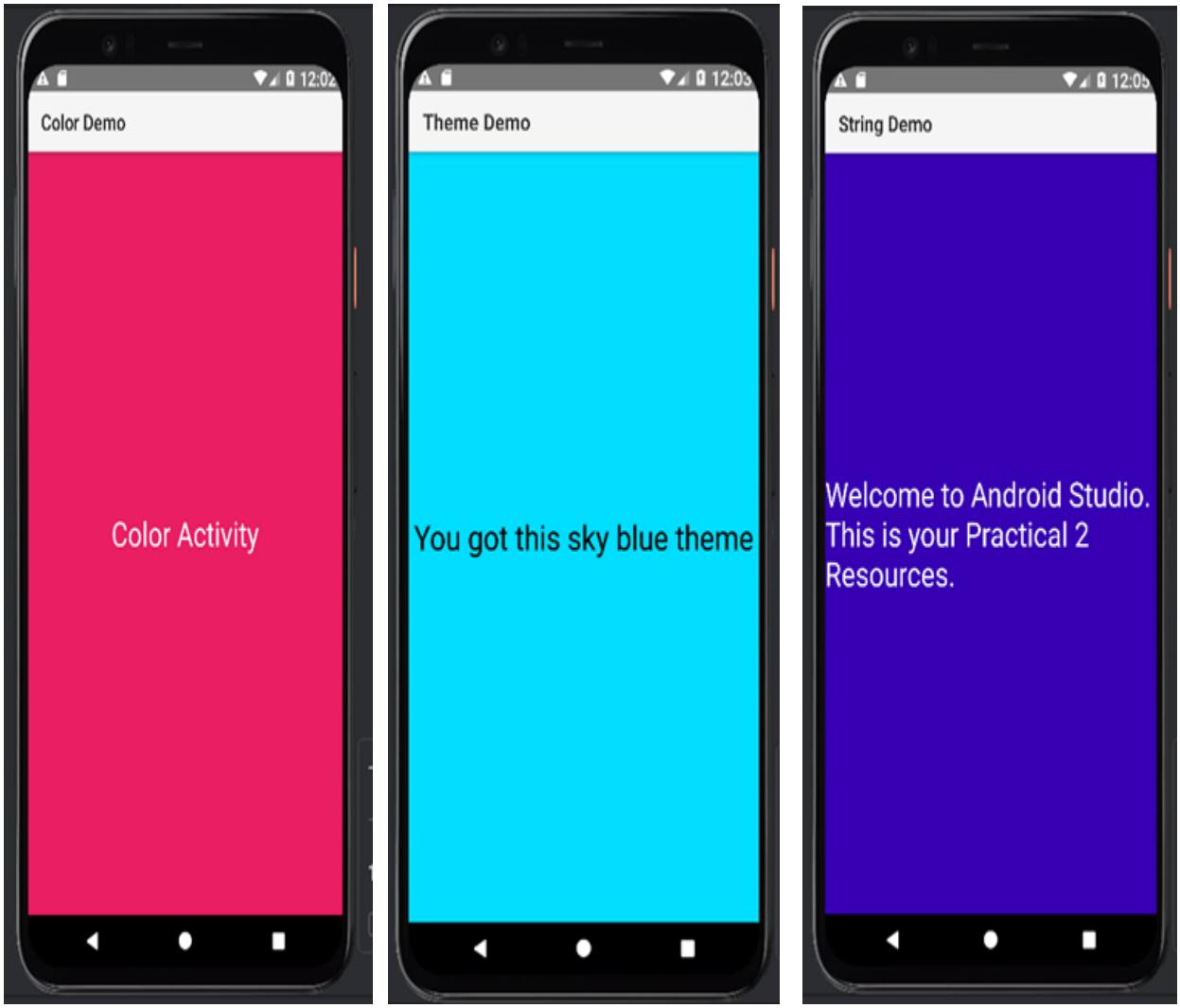


Output:

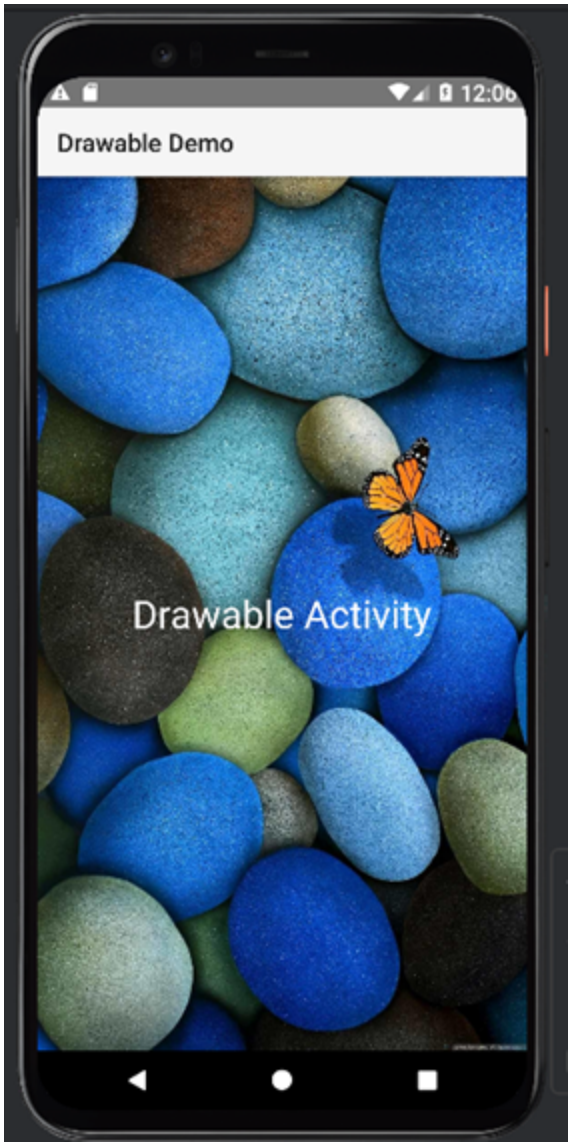
MAIN PAGE



Click COLOR DEMO press back & Click THEME DEMO press back & Click THEME DEMO



Press back & Click COLOR DEMO Press back & Click COLOR DEMO


12:06
Drawable Demo
Drawable Activity


12:06
Dimension Demo
Dimension Activity
CLICK ME!